

Final Report

Team 57

Abstract

This document is the final evaluation report presented by Team 57. This report includes the following sections in detail:

1. Progress Update
2. Strategy Overview
3. Performance Metrics
4. Unique Approaches and Innovations
5. Future Scope

We believe this report will help you understand our current strategy and help you evaluate it in detail.

1 Progress Update

We have currently employed two different strategies for BTC/USDT and ETH/USDT.

ETH- We used a combination of EMA, Smoothed RSI, Heiken Ashi Indicator, MACD, KST Indicator, Chaikin Volatility, Aroon Oscillator, Chebyshev Filter, Directional Movement Index (DMI), Percentage Change Indicator, Commodity Channel Index (CCI), Chande Momentum Oscillator and Donchian Channel for taking long and short trades.

BTC- We used a combination of EMA, Smoothed RSI, Heiken Ashi Indicator, MACD, KST Indicator, Chaikin Volatility, Aroon Oscillator, Elder Ray Index, Hawkes Process Directional Movement Index (DMI) and Percentage Change Indicator for taking long and short trades. We have de-noised using Chebyshev Filter and Kalman Filter.

2 Strategy on Bitcoin

2.1 Key Indicators and Filters Used:

- **EMA** : Exponential Moving Average
- **Smoothed RSI** : Smoothed Relative Strength Index
- **Heiken Ashi**

- **Hawkes Process**
- **MACD** : Moving Average Convergence Divergence
- **KST** : Know Sure Thing
- **Chaikin Volatility**
- **Aroon Oscillator**
- **Kalman Filter**
- **Chebyshev Filter**
- **DMI** : Directional Movement Index
- **Elder Ray Index**
- **Percentage Change Indicator**

2.1.1 Exponential Moving Average

The EMA is a moving average that places a greater weight and significance on the most recent data points. Like all moving averages, this technical indicator is used to produce buy and sell signals based on crossovers and divergences from the historical average.

$$EMA_t = \alpha \cdot Value_t + (1 - \alpha) \cdot EMA_{t-1}$$

where:

$$\alpha = \frac{\text{Smoothing}}{1 + \text{Days}}$$

and

EMA = Exponential Moving Average

2.1.2 Relative Strength Index

Relative Strength Index (RSI) is momentum based indicator used to conclude about the overbought or oversold nature of a commodity.

2.1.3 Smoothed RSI

Smoothed RSI is built upon RSI itself but is more efficient in capturing trend. Smoothed RSI applies the moving average procedure to RSI itself, making the indicator less twitchy and leading to fewer false positives.

2.1.4 Heikin Ashi

Heikin-Ashi is a candlestick pattern technique that aims to reduce some of the market noise, creating a chart that highlights trend direction better than typical candlestick charts. The Heikin-Ashi calculation creates a smoother-looking candlestick chart that makes it easier to identify and follow price trends.

$$HA\ Close = \frac{Open_t + High_t + Low_t + Close_t}{4}$$

$$HA\ Open = \frac{HA_{t-1} + HA_{t-1}}{2}$$

$$HA\ High = \max(High_t, HA\ Open_t, HA\ Close_t)$$

$$HA\ Low = \min(Low_t, HA\ Open_t, HA\ Close_t)$$

where:

- **Open_t, High_t, Low_t, Close_t** = Values from the current period
- **Open_{t-1}** = Values from the prior period
- **HA** = Heikin-Ashi

2.1.5 Moving Average Convergence Divergence

Moving average convergence/divergence (MACD) is a technical indicator to help investors identify entry points for buying or selling. The MACD line is calculated by subtracting the 26-period exponential moving average (EMA) from the 12-period EMA.

$$MACD = EMA_{12} - EMA_{26}$$

MACD signal line is also made using 9 period exponential moving average of the MACD value.

2.1.6 Know Sure Thing

Know Sure Thing (KST) is a momentum oscillator intended to interpret rate-of-change price data. KST signal line is calculated by taking the 9-period SMA of the KST value.

$$KST = (RMA_1) + 2(RMA_2) + 3(RMA_3) + 4(RMA_4)$$

where:

- **RMA₁** = 10-period SMA of 10-period ROC

- **RMA₂** = 10-period SMA of 15-period ROC

- **RMA₃** = 10-period SMA of 20-period ROC

- **RMA₄** = 15-period SMA of 30-period ROC

2.1.7 Chaikin Volatility

The Chaikin Volatility study is a technical indicator used to predict trend reversals based on volatility analysis. This indicator interprets volatility as expansion of the range between High and Low prices. As recommended by Chaikin, we are using 10 as the number of periods.

$$\overline{High-Low}_t = EMA_{10}(High_t - Low_t)$$

$$Volatility_t = \left(\frac{\overline{High-Low}_t - \overline{High-Low}_{t-10}}{\overline{High-Low}_{t-10}} \right) \times 100$$

2.1.8 Elder Ray Index

The Elder-Ray Index is a technical indicator that measures the amount of buying and selling pressure in a market. This indicator consists of two indicators known as "bull power" and "bear power," which are derived from a 13-period exponential moving average.

$$Bull\ Power = High_t - EMA_{13,t}$$

$$Bear\ Power = Low_t - EMA_{13,t}$$

2.1.9 Aroon Oscillator

The Aroon Oscillator measures the strength and direction of a trend. It is used in the strategy to confirm trend reversals, particularly for short exit signals when its value is above -10, indicating a potential weakening of a downtrend. Values that are positive imply an upward trend, whereas negative values signify a downward trend.

$$Aroon\ Oscillator = AU - AD$$

$$AU = \frac{n - \text{Periods Since Highest High}}{n} \cdot 100$$

$$AD = \frac{n - \text{Periods Since Lowest Low}}{n} \cdot 100$$

2.2 Directional Movement Index

The Directional Movement Index (DMI) is a technical tool that identifies the direction of an asset's price changes and assesses the trend's strength. It accomplishes this by contrasting the present price with prior peaks and troughs, producing two essential lines:

- Positive Directional Movement (+DI): Indicates the difference between today's highest price and yesterday's highest price, accumulated over the last 14 periods and graphed.
- Negative Directional Movement (-DI): Indicates the gap between today's low price and yesterday's low price, aggregated over the last 14 periods and illustrated.

In unstable markets, where trends may change quickly, the DMI successfully distinguishes between insignificant, random price fluctuations and real trends. In contrast to conventional indicators like moving averages that can produce misleading signals in volatile price movements, the DMI emphasizes the true market direction, assisting traders in managing market fluctuations.

H = High

L = Low

$$+DM = \begin{cases} H_t - H_{t-1}, & \text{if } (H_t - H_{t-1}) > (L_{t-1} - L_t) \\ & \text{and} \\ & (H_t - H_{t-1}) > 0, \\ 0, & \text{otherwise} \end{cases}$$

$$-DM = \begin{cases} L_{t-1} - L_t, & \text{if } (L_{t-1} - L_t) > (H_t - H_{t-1}) \\ & \text{and} \\ & (L_{t-1} - L_t) > 0, \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Smoothed } +DM = \text{EMA}(+DM)$$

$$\text{Smoothed } -DM = \text{EMA}(-DM)$$

$$+DI = 100 \cdot \frac{\text{Smoothed } +DM}{ATR}$$

$$-DI = 100 \cdot \frac{\text{Smoothed } -DM}{ATR}$$

2.2.1 Percentage Change Indicator

The rolling threshold is a statistical measure employed to determine the upper quantile of percentage variations across a rolling time window. This approach is used to establish a dynamic threshold for identifying significant price

fluctuations within the rolling window, aiding in the evaluation of possible exit points in trades.

$$\text{PctChange}[t] = \frac{\text{Close}[t] - \text{Close}[t - k]}{\text{Close}[t - k]} \times 100$$

$$\text{Rolling Threshold}[t] = Q_q(\text{PctChange}[t - w : t])$$

Q_q : Quantile function

k : Number of periods for percentage change.

Where: w : Size of the rolling window.

t : Current time step.

2.3 Denoisers and Filters

2.3.1 Kalman Filter

To filter out the inherent noisiness of financial markets filters are employed to better isolated underlying trends in the data. One such powerful tool is the Kalman Filter. It is a recursive algorithm that provides estimates of the true state of a system based on a series of noisy observations. It operates in two main steps

Prediction Step : In this step mathematical models of dynamical systems is used to predict the next state of the signal using the current state

Update Step : In this step, the Kalman Filter incorporates new measurements to update the state estimate. It calculates a weighted average of the predicted state and the new measurement, with weights determined by the respective uncertainties. The updated state estimate is more accurate because it takes into account both the prediction and the new information

2.3.2 Chebyshev Filter

The Chebyshev Filter smooths price data to minimize noise and reveal the underlying trend. It assists in eliminating abrupt price fluctuations, enabling entry or exit at more dependable signals. Its main characteristics consist of:

- **Order:** Establishes the filter's sharpness level

- **Ripple:** Manages the permitted ripple within the passband.

- **Cutoff Frequency:** Establishes the limit frequency for filtration

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$$H(s) = \frac{B(s)}{A(s)}$$

$$y[n] = \sum_{i=0}^M b_i x[n-i] - \sum_{j=1}^N a_j y[n-j]$$

2.4 Signal generation

In Bitcoin, the entry conditions for long and short trades are very strictly defined using a combination of the key indicators defined above but exit conditions are comparatively more relaxed for exiting a long and short trade. We will be looking into the logic in details below.

2.5 Long Entry

- $EMA_{20} \geq EMA_{50}$:- This is a simple bullish signal derived from an EMA Crossover, since EMA_{20} is more sensitive to price changes than EMA_{50} , a crossover means that bullish sentiment is rising and it may be a good time to enter into a long trade.
- $HA_{Open} \leq EMA_{20}$. This is a momentum confirmation signal. Since $HA_{Open} \leq EMA_{20}$, it confirms the trend set by EMA_{20} , which indicates a bullish trend according to the previous condition. It can be said that the market is treating EMA_{20} as support. This acts as confirmation.
- $HA_{Close} \geq EMA_{20}$. This is a momentum confirmation signal. Since $HA_{Close} \geq EMA_{20}$, it confirms the trend set by EMA_{20} , which indicates a bullish trend according to the previous condition. It can be said that the market is treating EMA_{20} as support. This acts as confirmation.
- Smoothed $RSI \geq 50$. RSI more than 50 acts as bullish trend confirmation.
- $Hawkes_{Close} \geq close$.
When this condition is activator, it indicates that predicted future price trends are aligning with or exceeding the current market price. This supports a bullish sentiment, suggesting a favorable opportunity to enter a long trade.
- Chaikin Volatility ≤ 25 :-
Explained in detail in 2.5.3.
- $MACD \leq MACD_{Signal}$:-
Explained in detail in 2.5.1.

- $KST \leq KST_{Signal}$. Explained in detail in 2.5.2.

- $Bull_{power} > 0$ & $Bear_{power} < 0$:-
The Elder Ray Index confirms a bullish trend.

- In the Directional Movement Index:

$$- +DI \geq -DI.$$

$$- +DI \geq 10.$$

A long buy signal is generated when all these conditions are met. Below, we have explained in detail our reasoning behind deviating from the conditions generally used in trading strategies. We have used the **mean-reversion counterpart** of the general trend-following approach.

2.5.1 MACD < Signal line

Condition that is generally used :

$$MACD > Signal\ Line$$

- Entering a long position earlier allows to take position before the potential shift in trend, capturing more of the uptrend if the anticipated momentum increase happens.
- Taking a long position when the MACD is below the Signal Line provides a better balance. The entry is based on early signs of momentum, and if the momentum doesn't continue, losses may be smaller since the position can be exited with limited downside. If waited until the MACD is above the Signal Line, the trend may already be well underway, and the potential upside may be more limited.
- Entering before the crossover allows to potentially capture the entire uptrend rather than a portion of it and allows to get even more profit.

2.5.2 KST < KST signal Line

Condition that is generally used :

$$KST > KST\ Signal\ Line$$

Entering a long position when the KST is less than the Signal Line for entering the market is preferred because of the same reasons as for Long entry with MACD, when MACD is less than the signal line

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- Entering a long position earlier allows to take position before the potential shift in trend, capturing more of the uptrend if the anticipated momentum increase happens.
- Taking a long position when the KST is below the Signal Line provides a better balance. The entry is based on early signs of momentum, and if the momentum doesn't continue, losses may be smaller since the position can be exited with limited downside. If waited until the KST is above the Signal Line, the trend may already be well underway, and the potential upside may be more limited.
- Entering before the crossover allows to potentially capture the entire uptrend rather than a portion of it and allows to get even more profit.

2.5.3 Chaikin volatility < 25

Condition that is generally used :

Chaikin Volatility > 30

When Chaikin Volatility is below a general threshold value, it indicates a period of low volatility. Entering a buy signal during low volatility can lead to profitable outcomes for following reasons:

- When Chaikin Volatility is low, it shows that the asset has been consolidating in a tight range and building momentum for a potential breakout.
- In a period of low volatility, prices tend to move within a narrow range. Entering in this interval gives a favourable entry price before a potential rise in price, rather than buying at a higher price after volatility increases.

2.6 Long exit

We have followed general conditions of long exit, without any particular deviations.

- $EMA_{20} < EMA_{50}$:-
This is a simple bearish signal derived from an EMA Crossover, since EMA_{20} is less sensitive to price changes than EMA_{50} , a crossover means that bearish sentiment is rising and it may be a good time to exit a long trade as close price may fall in the coming days.

- Kalman Filtered $<$ Closing Price:-
A lower filtered price compared to the closing price indicates downward momentum, supporting the choice to close the long position.
- $HA_{Open} > EMA_{20}$:-
The HA_{Open} staying above EMA_{20} reinforces the bearish trend, indicating that bullish momentum has decreased and prices may move against the position soon.
- $HA_{Close} < EMA_{20}$:-
Similar to the previous condition, this reinforces bearish sentiment, which supports other conditions to exit a long trade

The long buy signal is closed, and the trade has been exited when all these conditions are met.

2.7 Short Entry

- $EMA_6 < EMA_{10}$:-
We have used 6 time period and 10 time period, because we observed that Bitcoin shows short-lived downtrend. So in order to capture such trends, we used lower look-back period EMAs. This condition indicates that the shorter-term trend EMA_6 is below than the slightly longer-term trend EMA_{10} , signaling a shift in momentum to the downside. This bearish crossover suggests that prices are likely to continue falling, making it a good opportunity to enter a short trade.
- $HA_{Open} > EMA_6$:-
When $HA_{Open} > EMA_6$, it suggests that the market has experienced a recent downward move. This indicates bearish sentiment. So this might be a good opportunity to enter a short trade.
- $HA_{Close} < EMA_6$:-
This is also a trend following signal, similar to the previous condition.
- $MACD > MACD_{Signal}$:-
This is explained in detail in 2.7.1
- Chebyshev Filtered Price $<$ Close:-
This condition indicates that the current price is higher than the filtered or smoothed value, suggesting overbought

421	condition. This suggest an incoming price	466
422	correction. So this is a good signal for	467
423	entering a short trade.	468
424	• $RSI_{Smoothed} < 40$:- In general RSI be-	469
425	tween 25 and 45 indicates bearish senti-	470
426	ment. This further supports the previous	
427	bearish signals.	
428	• $+DI < -DI$:-	471
429	This condition indicates that the down-	472
430	ward price movement is stronger than the	473
431	upward movement, suggesting bearish sen-	474
432	timent. This is a signal that the price is	475
433	likely to continue decreasing, making it a	
434	good opportunity for a short trade.	
435	We have used $MACD > \text{Signal Line}$ instead of	
436	$MACD < \text{Signal line}$ as one of the short entry	
437	conditions. This is opposite of what is normally	
438	used in trend following approaches. We have	
439	explained our reasoning in detail below.	
440	2.7.1 $MACD > \text{Signal line}$	
441	Condition that is generally used :	
442	$MACD < \text{Signal Line}$	
443		
444	Entering a short position when the MACD is	
445	above the Signal Line for entering the market	
446	is preferred because of following:	
447	• Entering a short position earlier allows to	
448	take position before the potential shift in	
449	trend, capturing more of the downtrend.	
450	• Taking a short position when the MACD is	
451	above the Signal Line provides a better bal-	
452	ance. The entry is based on early signs of	
453	momentum, and if the momentum doesn't	
454	continue, losses may be smaller since the	
455	position can be exited with limited loss.	
456	Here we have used mean-reversion approach as	
457	opposed to the general trend following approach	
458	as previously explained in Long Entry	
459	2.8 Short Exit	
460	All general conditions have been applied with-	
461	out any deviations.	
462	• $EMA_6 > EMA_{10}$:-	476
463	We have used 6 time period and 10 time	477
464	period, because we observed that Bitcoin	478
465	shows short-lived downtrend. So in order	479
	to capture such trends, we used lower look-	480
	back period EMAs. This condition signals	481
	slight bullish sentiment in the market and	482
	is a good indicator of exiting the short	483
	trade before prices rise too much.	484
	• $HA_{Open} < EMA_6$:-	485
	This again confirms the previous indication	
	of bullish sentiment and is a good indicator	
	of exiting the short trade before prices rise	
	too much.	
	• $HA_{Close} > EMA_6$:-	486
	Similar to previous condition	487
	• $pct\ change_2 \leq \text{rolling threshold}$:-	488
	When the Percentage Change Indicator $\leq$	489
	Rolling Threshold, it suggests that the	490
	price movement is stabilizing or reversing,	491
	indicating a potential gain of upward mo-	492
	mentum. This is a good signal to exit a	
	short trade before the price turns against	
	the position.	
	• Aroon Oscillator > -10 :-	493
	When Aroon Oscillator is below 0, we con-	494
	sider the price to be in downtrend, but	495
	values close to 0, may signal a reversal of	496
	the current trend, or consolidation which	497
	is not ideal for our profit. Hence, this is a	498
	good signal to exit short trade.	499
	3 Strategy on Ethereum	500
	3.1 Key Indicators used	501
	• EMA : Exponential Moving Average	502
	• Heikin Ashi	503
	• MACD : Moving Average Convergence	504
	Divergence	505
	• Aroon Oscillator	506
	• Chaikin Volatility	
	• Chebyshev Filter	
	• Directional Movement Index (DMI)	
	• Smoothed RSI	
	• Commodity Channel Index (CCI)	
	• Chande Momentum Oscillator	
	• Donchian Channel	

The last 3 indicators in the list, listed above have not been defined before and are indicators that we have specifically introduced in this strategy. We have also not used Kalman filters for Ethereum, because of suboptimal results. We needed a sharp high pass filter here, to remove the extreme fluctuations and also eliminate signals generated from very short lived trends. So we used only Chebyshev filter here.

3.2 Insight into specific indicators used for Ethereum

3.2.1 Commodity Channel Index

The Commodity Channel Index (CCI) is a momentum-based oscillator that helps to detect overbought and oversold situations, along with possible reversals in price movements. It assesses the correlation between the present price, a moving average of that price, and the average deviation of price fluctuations.

$$CCI = \frac{\text{Typical Price} - \text{SMA}(\text{Typical Price}, n)}{0.015 \cdot \text{Mean Deviation}}$$

3.2.2 Chande Momentum Oscillator

The Chande momentum oscillator is a technical indicator that uses momentum to identify relative strength or weakness in a market. The chosen time frame greatly affects signals generated by the indicator. Pattern recognition often generates more reliable signals than absolute oscillator levels. Overbought-oversold indicators are less effective in strongly trending markets.

$$CMO = 100 \times \frac{\alpha - \beta}{\alpha + \beta}$$

where, α = Sum of Gains over n periods

β = Sum of Losses over n periods

3.2.3 Donchian Channel

Donchian Channels are a technical indicator that seeks to identify bullish and bearish extremes that favour reversals, higher and lower breakouts, breakdowns, and other emerging trends. The middle band computes the average between the highest high over a given period and the lowest low over the same time. These points identify the median or mean reversion price.

Upper Band = $\max(\text{Highs over } n \text{ periods})$

Lower Band = $\min(\text{Lows over } n \text{ periods})$

$$\text{Donchian Midline} = \frac{\text{Upper Band} + \text{Lower Band}}{2}$$

3.3 Identification of trades

3.3.1 Long Entry

- $EMA_{12} > EMA_{26}$:-

When the shorter-term EMA_{12} moves above the longer-term EMA_{26} , it suggests growing bullish momentum and a possible entry point for a long position.

- $HA_{\text{open}} < EMA_{20}$:-

The HA_{open} being lower than EMA_{20} validates bullish sentiment, supporting the initiation of a long position. This is also dependent on $HA_{\text{close}} > EMA_{20}$ being true, which is described in the next condition.

- $HA_{\text{close}} > EMA_{20}$:-

The HA_{close} above EMA_{20} strengthens the bullish trend and offers further validation for a long position.

- *Chaikin Volatility* < 55:-

Lower Chaikin volatility suggests diminished market fluctuations, indicating favorable conditions for starting a long trade.

When all these conditions are simultaneously satisfied a long position is opened with entry at the closing of current time period.

3.3.2 Stricter Long Entry

: (Triggered when holding time exceeds a threshold period) We hold for a maximum of 100 days. Since we are using 6 hr frequency candlestick, time calculator will count upto $100 \times (24/6) = 400$. This condition will only get checked if long trade exceeds 100 days.

- $EMA_{12} > EMA_{26}$:-

Similar to the typical long entry, this criterion validates positive momentum as the shorter EMA surpasses the longer EMA and signals bullish sentiment.

- $HA_{\text{open}} < EMA_{20}$ The Heiken Ashi opening price below EMA_{20} verifies that the

597 trend remains bullish, this also constrained
598 by the Heiken Ashi closing price condition
599 defined in the next bullet.

- 600 • $HA_{close} > EMA_{20}$ The Heiken Ashi clos-
601 ing price above EMA_{20} shows that there
602 is bullish sentiment. This condition will
603 always appear in pairs with it's opposite
604 condition involving Heiken Ashi open price
605 as described in the previous condition.
- 606 • $Chaikin\ Volatility < 50$ A stricter limit
607 for Chaikin volatility indicates the need
608 for more stable market conditions before
609 entering.
- 610 • $Smoothed\ RSI > 60$ A higher smoothed
611 RSI level at 60 ensures stronger bullish
612 momentum, meeting the stricter criteria
613 for entry.

614 3.3.3 Long exit

615 We have followed general conditions of long
616 exit, without any particular deviations.

- 617 • $EMA_{12} < EMA_{26}$:-
618 A bearish crossover of the shorter EMA
619 below the longer EMA suggests the con-
620 clusion of the bullish trend.
- 621 • $HA_{Open} > EMA_{20}$:-
622 The Heiken Ashi opening price being above
623 the EMA_{20} indicates declining bullish mo-
624 mentum, leading to an exit.
- 625 • $HA_{Close} < EMA_{20}$:-
626 The Heiken Ashi closing price below
627 EMA_{20} validates the bearish trend, sup-
628 porting the exit.
- 629 • $CMO_{i-1} \geq 20$:-
630 A Chande Momentum Oscillator value
631 above 20 indicates weakening upward mo-
632 mentum, reinforcing the decision to exit.

633 The long buy signal is closed, and the trade
634 has been exited when all these conditions are
635 met.

636 3.3.4 Short Entry

637 We have followed general conditions of Short
638 entry, without any particular deviations.

- 639 • $EMA_{12} < EMA_{26}$:-
640 A bearish crossover of the shorter EMA
641 below the longer EMA suggests the con-
642 clusion of the bullish trend.

- 643 • Chebyshev Filtered $< close$:-
644 A Chebyshev-filtered price below the clos-
645 ing price indicates bearish sentiment, sup-
646 porting a short entry.
- 647 • Minus $DI_s < Plus\ DI_s$:-
648 The $-DI$ surpassing $+DI$ indicates an
649 emerging bearish trend, reinforcing the en-
650 try decision.
- 651 • Minus $DI_s > 0$:-
652 A positive $-DI$ value suggests existing bear-
653 ish momentum, validating the short entry.

654 When all the above stated are simultaneously
655 satisfied, then a short trade is entered.

656 3.3.5 Strict Short Entry

657 : (Triggered when holding time exceeds a
658 threshold period) We hold for a maximum of
659 100 days. Since we are using 6 hr frequency
660 candlestick, time calculator will count upto
661 $100 \times (24/6) = 400$. This condition will only get
662 checked if short trade exceeds 100 days.

- 663 • $EMA_{12} < EMA_{26}$:-
664 Similar to the typical short entry, this cri-
665 terion confirms bearish momentum as the
666 shorter EMA drops below the longer EMA.
- 667 • Chebyshev Filtered $< close$:-
668 A lower Chebyshev-filtered price confirms
669 persistent bearish sentiment for a stricter
670 entry.
- 671 • Minus DI_s is $> Plus\ DI_s$:-
672 A stronger bearish trend is validated when
673 $-DI$ surpasses $+DI$.
- 674 • Smoothed RSI is < 30 :-
675 A smoothed RSI below 30 signifies oversold
676 conditions, aligning with stringent bearish
677 entry requirements.

678 3.3.6 Short Exit

679 We have followed general conditions of Short
680 exit, without any particular deviations.

- 681 • CCI is greater than -20 :-
682 This signal indicates declining bearish mo-
683 mentum and a potential reversal, prompt-
684 ing an exit.
- 685 • Aroon Oscillator > -50 :-
686 n Aroon Oscillator above -50 signals weak-
687 ening bearish momentum, supporting the
688 exit.

- $CMO_{i-1} \leq 30$:-
A Chande Momentum Oscillator value below 30 confirms diminishing bearish momentum, suggesting an exit point.
- $close > Donchian\ Middle_{i-1}$:-
The closing price exceeding the middle Donchian channel indicates a trend change, emphasizing the need to exit the short position.
- $pct\ change_2 \leq Rolling\ Threshold$:-
This indicates that the price change has slowed down and is within an acceptable range, suggesting that bearish momentum is losing strength. As a result, it becomes a signal to exit the short trade, as the market may be reversing or consolidating. The acceptable range is defined by selecting a percentile

When all the conditions are satisfied simultaneously, then the short signal is closed, and the trade has been exited.

3.4 Timeout exit

If the holding time of long trade reaches 100 days, then that particular trade is closed and stricter long entry conditions are checked on the immediate next period. If the holding time of short trade reaches 100 days, then that particular trade is closed and stricter short entry conditions are checked on immediate next interval

- Time calculator > 400 .

if time calculator reaches 100 days, then long/short trade is immediately exited.

3.5 Unique Appraoch and Innovation

In the entire duration of developing a robust strategy for bitcoin and ethereum, we started out by checking the performance of each indicator in addition to the prior indicator. We started with EMA, then moved on to RSI, Bollinger bands, ADX, ATR, then checking strategies based on trend, volatility, volume, statistics, etc. We found that BTC is a more stable, upward moving cryptocurrency while Ethereum is a highly volatile, longer-lived down-trend cryptocurrency. In BTC we created strict conditions for entering a long or short trade since trend is not going to change rapidly and

created more easy to exit strategies as those signals would also be strong enough to make a judgement. However, in ETH we found that even if we entered a long/short trade, the trend soon changes, in a short span of time. So, we decided that we will make it easy to enter but will check more conditions for exiting the trade. This turned out to be a robust strategy as we entered the trade, long or short, and if signals changed or it turned out to be noise, then we would exit the trade. If we created even more restrictive or harsh conditions on entering a trade, then number of trades fall significantly, which is not good for a robust strategy. We have also deviated from general usage of MACD, KST indicator and Chaikin volatility, in BTC. We used filtering signals like chebyshev Filtering which are used in signals processing for filtering noise in BTC and ETH. We also figured out that volume based indicators will give poor performance since cryptocurrencies are traded across hundreds of exchanges globally. Each exchange has its own trading volume data, making it difficult to get a comprehensive and reliable measure of total trading volume.

4 Results

In this section we provide the analysis of the backtesting results of both the strategies discussed above. Backtesting has been conducted on 18 quarters starting from 08-09-2019 to 01-01-2024, leverage of 1 is applied and both long and short trades have been taken on BTC as well as ETH.

4.1 Etherium

Table 1 highlights the results of our backtesting on Etherium. With a compounded total returns of 18084.8% we have succesfully beat the benchmark return of 1442.28%. The strategy has managed to keep the drawdown limited of 25.69% at maximum. With sharpe of 4.93, sortino of 54.47, an annual excess return (alpha) of 134.2 % and a maximum adverse excursion of 36.89 the strategy has given great returns for the risk associated with the same.

Figure 1 highlights the points at which trade entry signals were triggered for both long and short positions. Figure 2 shows how the compounded portfolio changes with closing of each trade. Figure 3 shows how the compounded

Metric	Result
Initial Portfolio	\$ 1000
Leverage	1
No. of Trades	93
Compunded Final Portfolio	\$ 181848.11
Static Final Portfolio	\$ 8932.21
Strategy Return	18084.8 %
Benchmark Return	1442.28 %
Win Rate	45.16 %
Maximum Drawdown	25.69 %
Time to Recovery	145 Days
Maximum Holding Time	100 Days
Alpha Ratio	134.2 %
Sharpe Ratio	4.93
Sortino Ratio	54.47
Maximum Adverse Excursion	36.86

Table 1: Details regarding backtesting for strategy on Ethereum.

portfolio changes over time with trades being taken and closed with time.

4.2 Bitcoin

Table 2 details the results of our backtesting conducted on the strategy built on BTC data. The strategy has achieved a final compounded portfolio value of \$73619.04 with a maximum drawdown of 10.91%. Sharpe and sortino are at 6.159 and 140.47 respectively while giving an excess return of 122% when compared to the benchmark. The returns generated per trade are also balanced hence solidifying the robust nature of the strategy. It shows displays a good mix of long and short trades and a decent win rate as well.

Figure 4 points out the timestamps at which entry signals were triggered for both long and short positions. Figure 5 shows how the compounded portfolio changes with closing of each trade. Figure 6 shows how the compounded portfolio changes over time with trades being taken and closed with time.

5 Future Scope

5.1 Bitcoin

We have acheived appreciably low drawdown levels and balanced trades on bitcoin. We plan to expand our strategy to higher leverages and maintain or improve its statistics

Metric	Result
Initial Portfolio	\$ 1000
Leverage	1
No. of Trades	55
Compunded Final Portfolio	\$ 73619.03
Static Final Portfolio	\$ 6944.20
Strategy Return	7261.90 %
Benchmark Return	326.47 %
Win Rate	52.73 %
Maximum Drawdown	10.91 %
Time to Recovery	136.17 Days
Maximum Holding Time	123 Days
Alpha Ratio	122 %
Sharpe Ratio	6.159
Sortino Ratio	140.47
Maximum Adverse Excursion	32.66

Table 2: Details regarding backtesting for strategy on Bitcoin.



Figure 1: Trade Initiations on ETH

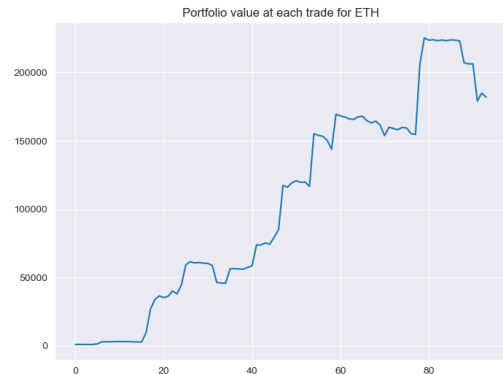


Figure 2: Portfolio of ETH with trades

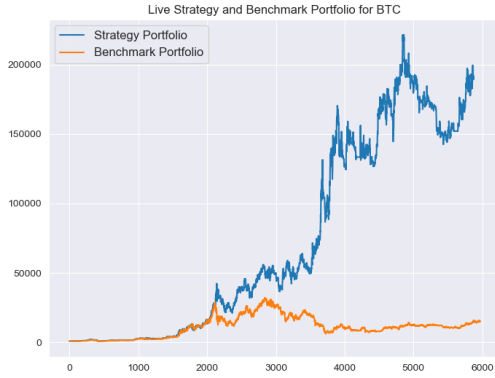


Figure 3: Live Portfolio Values for ETH



Figure 4: Trade Initiations on BTC

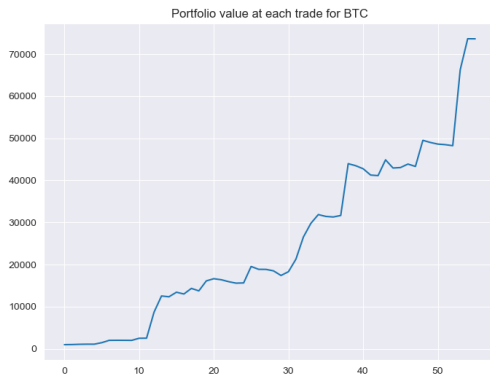


Figure 5: Portfolio of BTC with trades



Figure 6: Live Portfolio Values for BTC

1. Adding more indicators to increase the robustness of the strategy and decrease false positives. 813
2. Identification of false positives using machine learning algorithms and indicators to exit early from loss making positions 816
3. Improving our risk management strategy by implementing more robust continuation signal identifiers. 819

5.2 Ethereum 822

We have noticed that our strategy has strong trend identification properties we plan to build upon that and employ risk management strategies to decrease drawdown to even lower values. We have also taken lower amounts of long positions which we plan to address as well. 823

1. Adding weak trend continuation indicators to identify weak trends while trade is active 829
2. Loosening long entry conditions while ensuring minimal false positives to successfully exploit long trade opportunities 831
3. Anomaly detection using fourier transform of volume and LSTM based approach to predict the same 834